

statements are correct.) [Check All as answer]

1. Teaching how to use Emacs for personal productivity
2. Acting as innovation coach for those building a startup
3. Working on augmenting AI to enhance Karmic Design Thinking
4. Conducting online baking classes

Select the correct answer from the options below:

Options :

6406533516920. ✖ Only 1

6406533516921. ✖ Only 2

6406533516922. ✖ Only 3

6406533516923. ✔ All

Game Theory

Section Id :	64065375431
Section Number :	15
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653160264
Question Shuffling Allowed :	No

Question Number : 348 Question Id : 6406531041088 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533516924. ✓ YES

6406533516925. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653160265

Question Shuffling Allowed :

No

Question Id : 6406531041089 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (349 to 354)

Question Label : Comprehension

Consider the market for a single network good as discussed in the lecture and write a different consumer's utility function for joining the network as $U(\theta) = \theta(a + vn^e)$ capturing the idea that it may be more plausible that a user who has a higher value for the standalone component of a technology also assigns more importance to the size of its network (i.e. consumers differ in their valuation of both the standalone and the network benefits). Here, a is the standalone benefit, $v > 0$ measures the network effect, n^e is the expected number of users joining the network and θ is uniformly distributed on the unit interval.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 349 Question Id : 6406531041090 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

The value $\hat{\theta}$ corresponding to an indifferent consumer for a given price p and a given expected network size n^e will be

Options :

6406533516926. ✗ $\frac{p-a}{vn^e}$

6406533516927. ✓ $\frac{p}{(a+vn^e)}$

6406533516928. ✗

$$\frac{ap}{(a+vn^e)}$$

6406533516929. ✖ None

Question Number : 350 Question Id : 6406531041091 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The maximum value of p to have buyers will be

Options :

6406533516930. ✔ $a + vn^e$

6406533516931. ✖ vn^e

6406533516932. ✖ $\frac{a}{vn^e}$

6406533516933. ✖ $\frac{a}{1+vn^e}$

Question Number : 351 Question Id : 6406531041092 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The mass of buyers (n) as a function of $\hat{\theta}$ will be

Options :

6406533516934. ✖ $\hat{\theta}$

6406533516935. ✖ $1 + \hat{\theta}$

6406533516936. ✔ $1 - \hat{\theta}$

6406533516937. ✖ $\frac{1+\hat{\theta}}{1-\hat{\theta}}$

Question Number : 352 Question Id : 6406531041093 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The willingness to pay ($p(n^e, n)$) for the n^{th} unit of the good when n^e units are expected to be sold will be

Options :

6406533516938. ✖ $(a + vn^e)$

6406533516939. ✖ $n(a + vn^e)$

6406533516940. ✖ $(a + vn^e)(1 + n)$

6406533516941. ✔ $(a + vn^e)(1 - n)$

Question Number : 353 Question Id : 6406531041094 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

$p(n^e, n)$ is

Options :

6406533516942. ✖ increasing function of n

6406533516943. ✔ decreasing function of n

6406533516944. ✖ decreasing function of n^e

6406533516945. ✔ increasing function of n^e

Question Number : 354 Question Id : 6406531041095 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The fulfilled-expectations demand curve in this setting will be

Options :

6406533516946. ✖ $(a + vn)$

6406533516947. ✖ $n(a + vn)$

6406533516948. ✖ $(a + vn)(1 + n)$

6406533516949. ✓ $(a + vn)(1 - n)$

Sub-Section Number : 3
Sub-Section Id : 640653160266
Question Shuffling Allowed : No

Question Id : 6406531041108 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (355 to 357)

Question Label : Comprehension

Calculate the Shapley value as the average of players' Marginal contributions for the following game:

$$N = \{1, 2, 3\}$$

$$v(1) = 6 \quad v(2) = 12 \quad v(3) = 18 \quad v(\phi) = 0$$

$$v(1, 2) = 12 \quad v(1, 3) = 12 \quad v(2, 3) = 24$$

$$v(1, 2, 3) = 48$$

Based on the above data, answer the given subquestions

Sub questions

Question Number : 355 Question Id : 6406531041109 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

$$S_1 = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Question Number : 356 Question Id : 6406531041110 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

$$S_2 = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

14

Question Number : 357 **Question Id :** 6406531041111 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

$S_3 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

23

Sub-Section Number : 4

Sub-Section Id : 640653160267

Question Shuffling Allowed : No

Question Id : 6406531041122 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (358 to 360)

Question Label : Comprehension

There are 3 firms that are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$ where n is the number of firms that enter.

The cost of entering is 62.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 358 **Question Id :** 6406531041123 **Question Type :** MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

How many pure strategy equilibria are there in this game?

Options :

6406533516988. ✖ 1

6406533516989. ✖ 2

6406533516990. ✔ 3

6406533516991. ✖ 4

Question Number : 359 Question Id : 6406531041124 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In one of these pure NEs, all three players enter the market

Options :

6406533516992. ✖ TRUE

6406533516993. ✔ FALSE

Question Number : 360 Question Id : 6406531041125 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

There is a symmetric mixed strategy equilibrium where all three players enter with the same probability (say p). Find the value of p

Options :

6406533516994. ✖ 1/5

6406533516995. ✖ 3/5

6406533516996. ✔ 4/5

6406533516997. ✖ 2/5

Sub-Section Number :

5

Sub-Section Id :

640653160268

Question Shuffling Allowed :

No

Question Id : 6406531041099 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (361 to 368)

Question Label : Comprehension

A market game has two buyers and one seller. Seller 1 has a watch whose worth for him is ₹200 and he offers it for sale. Players 2 and 3 are two potential buyers who do not own a watch. The worth of watch is ₹280 for one buyer (player 2) and ₹300 for the other buyer (player 3). Each buyer is interested in paying the lowest price for the watch and unwilling to pay more than the watch's worth to him. Describe the game in coalition form and answer the given sub questions.

Sub questions

Question Number : 361 Question Id : 6406531041100 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(1) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

200

Question Number : 362 Question Id : 6406531041101 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(2) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 363 Question Id : 6406531041102 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(3) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 364 Question Id : 6406531041103 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$$v(\phi) = \underline{\hspace{2cm}}$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 365 Question Id : 6406531041104 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(2, 3) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 366 Question Id : 6406531041105 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(1, 2) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

280

Question Number : 367 Question Id : 6406531041106 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(1, 3) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Question Number : 368 Question Id : 6406531041107 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

$v(1, 2, 3) = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

300

Sub-Section Number : 6

Sub-Section Id : 640653160269

Question Shuffling Allowed : No

Question Id : 6406531041096 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (369 to 370)

Question Label : Comprehension

Suppose that you receive a piece of e-mail whose subject line contains the phrase “check this out” (a popular phrase among scammers). Based just on this and without looking at the sender or the message content, what is the chance that the message is spam? Suppose that 40% of all your emails are spam and the remaining 60% are emails you want to receive. Suppose that 1% of all spam messages contain the phrase “check this out” in their subject lines, while 0.4% of all non-spam messages contain this phrase.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 369 Question Id : 6406531041097 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the total probability of having the phrase “check this out” in subject line of an email?

Options :

6406533516950. ✖ 0.64

6406533516951. ✖ 0.064

6406533516952. ✔ 0.0064

6406533516953. ✖ 0.00064

Question Number : 370 Question Id : 6406531041098 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the probability that the message is spam if subject contains the phrase "check this out"?

Options :

6406533516954. ✓ 0.625

6406533516955. ✗ 0.0625

6406533516956. ✗ 0.00625

6406533516957. ✗ 0.000625

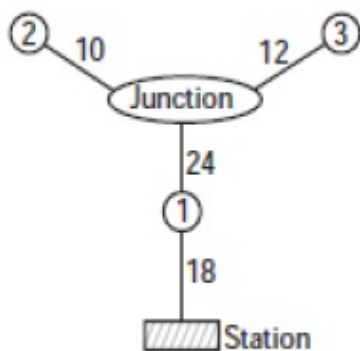
Question Id : 6406531041112 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (371 to 373)

Question Label : Comprehension

A cable TV station wants to connect three new customers to its network. The connections to the network are described in the following tree graph:



Calculate the distribution of the costs among the three players if all three decide to connect to the network.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 371 Question Id : 6406531041113 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 1 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 372 Question Id : 6406531041114 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 2 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

28

Question Number : 373 Question Id : 6406531041115 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 3 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Id : 6406531041116 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (374 to 375)

Question Label : Comprehension

Five bidders with private valuations uniformly distributed between 0 and 1 participate in a sealed bid first price auction. Assuming that everyone is playing the symmetric equilibrium bidding strategy answer the given subquestions:

Sub questions

Question Number : 374 Question Id : 6406531041117 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Optimal bid for a bidder with valuation 0.75 will be

Options :

6406533516972. ✖ 0.7

6406533516973. ✔ 0.6

6406533516974. ✖ 0.5

6406533516975. ✖ 0.75

Question Number : 375 Question Id : 6406531041118 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

The expected revenue of the seller in this auction scenario will be

Options :

6406533516976. ✖ 4/5

6406533516977. ✖ 3/5

6406533516978. ✔ 2/3

6406533516979. ✖ 2/5

Question Id : 6406531041119 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (376 to 377)

Question Label : Comprehension

Synergies: Two division managers can invest time and effort in creating a better working relationship. Each invests $e_i \geq 0$, and if both invest more then both are better off, but it is costly for each manager to invest. In particular, the payoff function for player i from effort levels (e_i, e_j) is $v_i(e_i, e_j) = (a + e_j)e_i - e_i^2$.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 376 Question Id : 6406531041120 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Best response correspondence for player i , $BR_i(e_j)$, will be

Options :

6406533516980. ✖ $\frac{a+2e_j}{2}$

6406533516981. ✔ $\frac{a+e_j}{2}$

6406533516982. ✖ $a + e_j$

6406533516983. ✖ $\frac{a+2e_j}{3}$

Question Number : 377 Question Id : 6406531041121 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Nash equilibrium of this game will be

Options :

6406533516984. ✖ $e_1 = e_2 = \frac{a}{2}$

6406533516985. ✖ $e_1 = e_2 = 0$

6406533516986. ✖ $e_1 = e_2 = \frac{a}{3}$

6406533516987. ✔ $e_1 = e_2 = a$

Question Id : 6406531041129 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (378 to 380)

Question Label : Comprehension

Suppose a search engine has two ad slots that it can sell. Slot **a** has a clickthrough rate of 10 and slot **b** has a clickthrough rate of 5. There are three advertisers who are interested in these slots. Advertiser **x** values click at 3 per click, advertiser **y** values click at 2 per click, and advertiser **z** values click at 1 per click.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 378 Question Id : 6406531041130 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Total valuation of socially optimal allocation will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

40

Question Number : 379 Question Id : 6406531041131 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

VCG price for slot **a** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 380 Question Id : 6406531041132 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

VCG price for slot **b** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :

7

Sub-Section Id :

640653160270

Question Shuffling Allowed :

No

Question Id : 6406531041126 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (381 to 382)

Question Label : Comprehension

A man dies, leaving an estate worth 500 units. The deceased has two creditors; one's claim is 400 units, and the other's claim, 300 units. The division of the estate between them is carried out according to the contested garment principle.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 381 Question Id : 6406531041127 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Creditor 1 with claim 400 units gets

Options :

6406533516998. ✖ 350

6406533516999. ✔ 300

6406533517000. ✖ 200

6406533517001. ✖ 250

Question Number : 382 Question Id : 6406531041128 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Creditor 2 with claim 300 units gets

Options :

6406533517002. ✖ 150

6406533517003. ✖ 300

6406533517004. ✔ 200

6406533517005. ✖ 250

Sub-Section Number :

8

Sub-Section Id :

640653160271

Question Shuffling Allowed :

Yes

Question Number : 383 Question Id : 6406531041133 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Suppose 5 numbers are drawn independently from the uniform distribution on the interval $[0, 1]$ and then sorted from smallest to largest. The expected value of the number in second position on the list is

Options :

6406533517009. ✖ 5/6

6406533517010. ✖ 2/3

6406533517011. ✖ 1/2

6406533517012. ✔ 1/3

Sub-Section Number :

9

Sub-Section Id :

640653160272

Question Shuffling Allowed :

Yes

Question Number : 384 Question Id : 6406531041134 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In weighted majority game $[7; 5, 4, 3, 1]$

Options :

6406533517013. ✖ Player 1 is null player

6406533517014. ✖ Player 2 is null player

6406533517015. ✖ Player 3 is null player

6406533517016. ✔ Player 4 is null player

Question Number : 385 Question Id : 6406531041135 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider following two statements:

1. Crowd can be wrong even if everyone in the crowd is rational and everyone takes the same action.
2. Aggregate behaviour of many people with limited information can sometimes produce very accurate results.

Which statement(s) is (are) true?

Options :

6406533517017. ✖ Only 1

6406533517018. ✖ Only 2

6406533517019. ✔ Both 1 and 2

6406533517020. ✖ None of these

Question Number : 386 Question Id : 6406531041136 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406533517021. ✖ A stable matching system is necessarily a system under which everyone is satisfied.

6406533517022. ✖ A matching system is stable when an unmatched pair may find it beneficial to deviate from the matching and form their own match.

6406533517023. ✖ A stable matching system is necessarily a system under which everyone is satisfied & A matching system is stable when an unmatched pair may find it beneficial to deviate from the matching and form their own match.

6406533517024. ✔ None of these